1. **In the forward diffusion process, what happens to an image as t increases toward the final timestep?** 1 point

- All noise is added at once in the final timestep.
- Noise is gradually removed so the image becomes sharper.
- Noise is added incrementally until the image becomes nearly pure noise.
- Pixels are randomly shuffled but their values stay the same.

2. **A max-pool layer with kernel = 2, stride = 2 is inserted after a convolution in a CNN. What effect does this layer have on the feature tensor?** 1 point

- Keeps channel count unchanged and halves both spatial dimensions.
- Halves the channels and doubles both height and width.
- Keeps spatial dimensions unchanged but reduces information density.
- Doubles the number of channels and halves height × width.

3. **Saliency maps and Grad-CAM both rely on gradients, yet they often highlight different regions. Why?** 1 point

- Saliency computes pixel-level gradients; Grad-CAM aggregates over feature map regions.
- Grad-CAM filters out negative gradient contributions before visualization.
- Saliency uses gradients from all layers; Grad-CAM only from early layers.
- They compute gradients with respect to different loss functions.

4. **A CNN includes the sequence Conv (3×3, stride 1) ⟶ MaxPool (2×2, stride 2).** 1 point

If an engineer swaps their order (MaxPool first, Conv second) while keeping kernel sizes and strides unchanged, what is the most direct consequence for the effective receptive field of the conv layer?

- It becomes undefined because convolution cannot follow pooling.
- It is cut in half because pooling removed information the conv could use.
- It doubles in each spatial dimension because pooling was moved earlier.
- It stays 3×3 since the kernel size is unchanged.

5. **When generating an image with Stable Diffusion, increasing num_inference_steps from 20 to 50 while leaving all other parameters constant will most likely:** 1 point

- Increase variety and randomness in the generated output.
- Produce sharper, higher-fidelity images at the cost of longer runtime.
- Cause the image to become over-processed and lose natural details.
- Make the model follow the text prompt more literally and strictly.

6. **What is the purpose of the *negative_prompt* argument in Stable Diffusion pipelines?** 1 point

- It generates the opposite of what the main prompt describes.
- It reduces the overall strength of the main prompt's influence.
- It provides a safety filter that blocks inappropriate content.
- It tells the model what content to avoid.

7. **A diffusion model takes a noisy image and a timestep as input. What does it predict as output?** 1 point

- The total noise present in the current image.
- A small amount of noise to remove from the current image.
- What the image looked like one step earlier in the process.
- What the final clean image will look like.

8. **Why do diffusion models generate images through many small denoising steps instead of removing all the predicted noise at once?** 1 point

- Multiple steps are required for the mathematics to work correctly.
- The model can only predict a small amount of noise at a time
- The model's predictions aren't perfect, so one big correction produces poor results.
- Removing all noise at once would take too much computational power.

9. **A CNN uses 3×3 filters throughout its architecture. How can it recognize both tiny local edges and large overall shapes when the filter size never changes?** 1 point

- The network learns to ignore small details in deeper layers.
- The model uses different filter sizes for different parts of the image.
- Deeper layers use larger filters even though they're still called 3×3.
- Each 3×3 filter covers a larger input region in deeper layers.

10. **Suppose a Grad-CAM heatmap for class "zebra" highlights background grass instead of the striped animal. Which diagnosis best fits this outcome?** 1 point

- The last conv layer's receptive field is too small to see the zebra.
- The gradient calculation needs more denoising steps to converge properly.
- The classifier learned to associate grass patterns with zebra predictions.
- The image needs to be cropped so the zebra is centered before computing CAM.